



FABRIC STRUCTURE & DESIGN

Part: B

Course Code: FE 305-0723

Double Cloths

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STITCHED DOUBLE CLOTH

Stitched double cloth is made of two separate fabric layers:

Face cloth = Top layer

Back cloth = Bottom layer

Each layer has its own warp and weft yarns. *The two layers are then connected or stitched together.*

Purpose:

It can be used to improve *thickness, warmth, strength*, appearance, color effects, and design effects.

Classification:

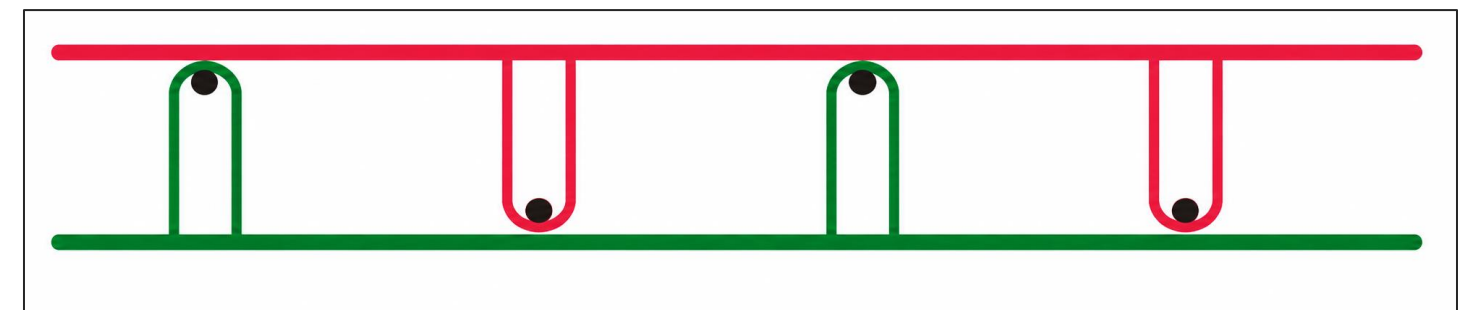
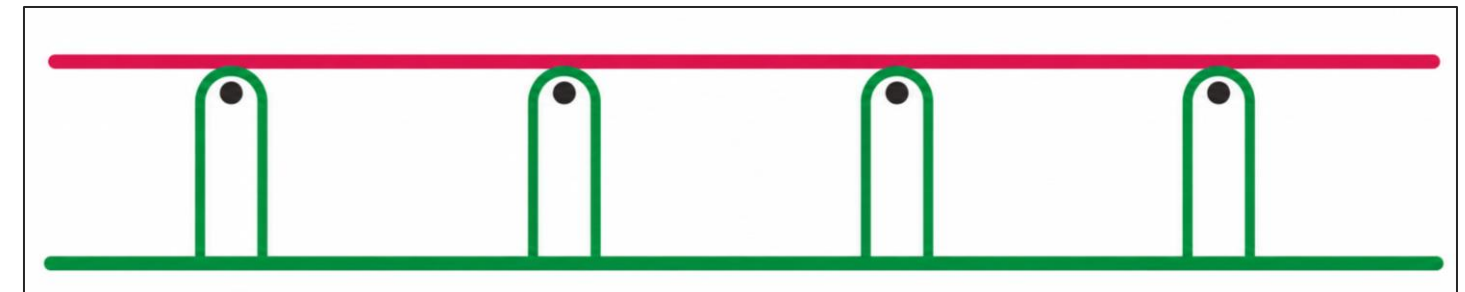
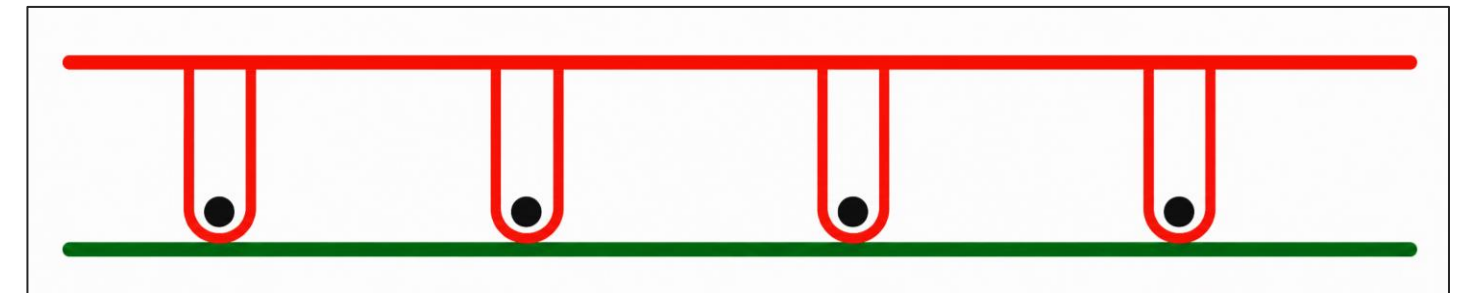
1. Self-Stitched Double Cloth.
2. Centre-Stitched Double Cloth.
3. Double Cloth Stitched by Thread Interchange.
4. Double Cloth Stitched by Cloth Interchange.
5. Alternate Single-Ply and Double-Ply Construction.

CLASSIFICATION

1. Self-Stitched Double Cloth

- *No extra stitching yarn* is used.
- The face and back layers stitch each other.
- Stitching can be done by:
 - ❖ Dropping a face warp under a back weft, or
 - ❖ Lifting a back warp over a face weft, or
 - ❖ Using both methods.

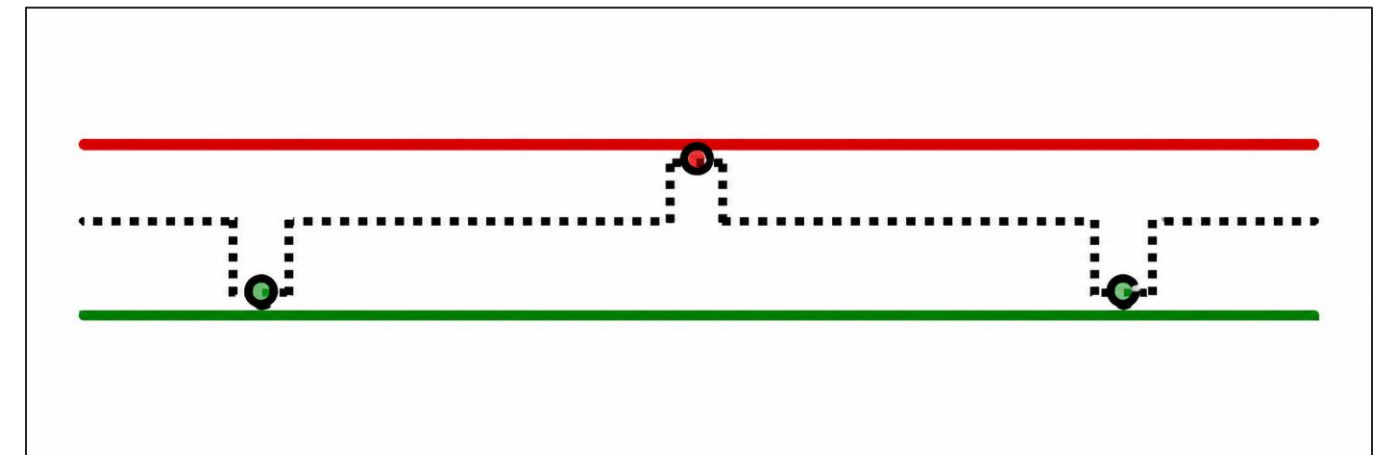
Easy idea: Face and back yarns themselves join the two layers.



CLASSIFICATION

2. Centre-Stitched Double Cloth

- *A third set of yarns is added* only for stitching.
- These stitching yarns stay between the face and back layers.
- They connect the two layers at *regular intervals*.

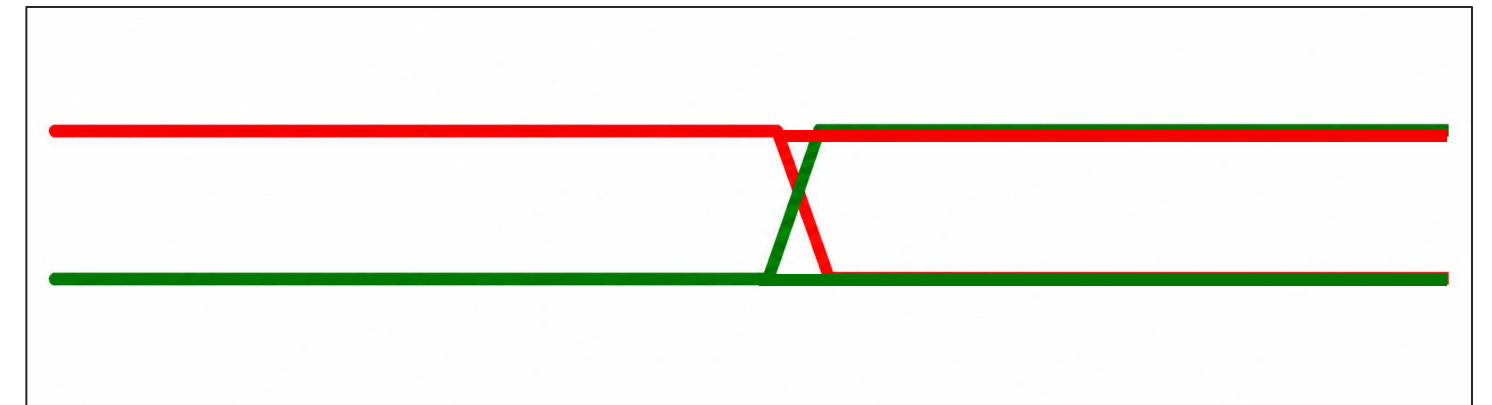


Easy idea: An extra middle yarn joins the face and back fabrics.

CLASSIFICATION

3. Double Cloth Stitched by Thread Interchange

- No extra stitching yarn is needed.
- Some face and back yarns change between the two layers.
- Face warp may interweave with back weft.
- Back warp may interweave with face weft.
- The point where they interchange is called the stitch point.



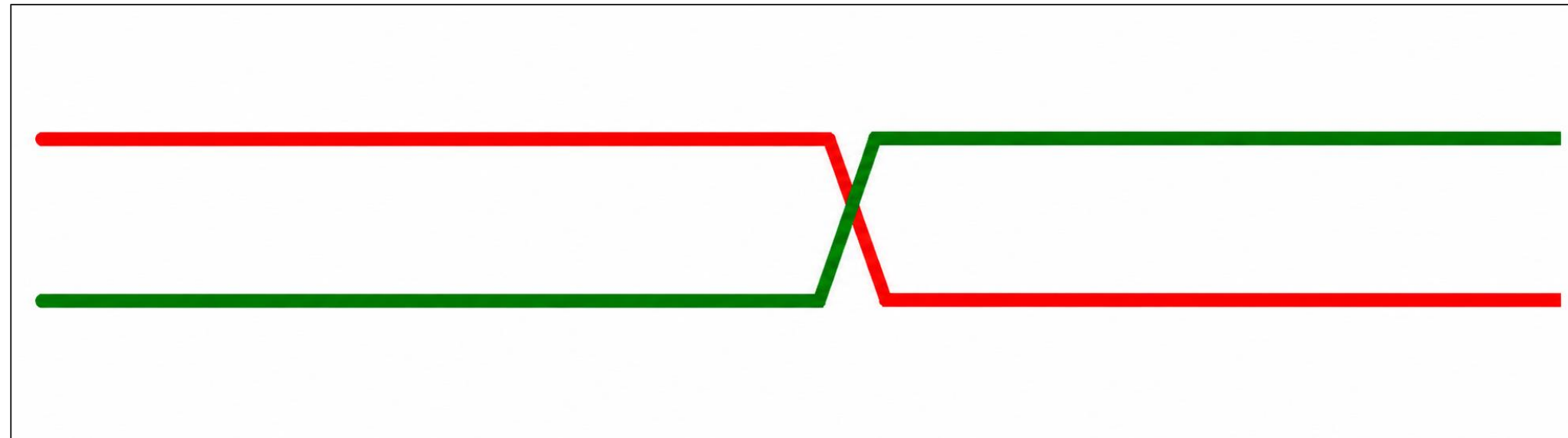
Easy idea: Some yarns move from one layer to the other to join them.

CLASSIFICATION

4. Double Cloth Stitched by Cloth Interchange

- The whole face and back layers change places.
- Stitching happens at the points where the layers interchange.
- More interchange points give stronger joining.

Easy idea: The complete upper and lower fabrics cross and exchange positions.

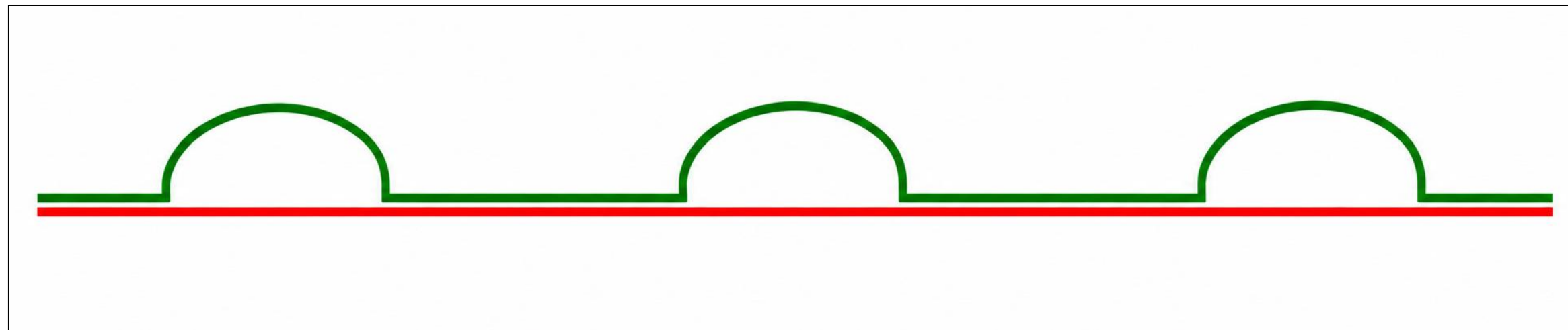


CLASSIFICATION

5. Alternate Single-Ply and Double-Ply Construction

- In some areas, the two layers are *joined into one single thick fabric*.
- In other areas, *they remain separate* as double cloth.
- The *separated parts* may form *pockets or raised* effects.

Easy idea: Some parts are single layer, while other parts remain double layer.



CONSIDERING POINTS BEFORE DOUBLE CLOTH CONSTRUCTION

1. Relative Proportion and Thickness of the Face & Back Threads:

❑ Thread Arrangement in Double Cloth:

- ✓ The arrangement of face and back threads mainly depends on the weight of the face texture.
- ✓ The order of weft threads is also influenced by the weft insertion system of the loom.

❑ Common Arrangements of Double Cloth:

❖ The most common arrangements are:

1 Face : 1 Back (1F:1B)

2 Face : 1 Back (2F:1B)

❖ For looms with a single-side box and when the back weft is different from the face weft, similar effects can be produced by using:

2 Face : 2 Back (2F:2B)

4 Face : 2 Back (4F:2B)

CONSIDERING POINTS BEFORE DOUBLE CLOTH CONSTRUCTION

- ❖ For fabrics requiring a **very fine face surface**, the arrangement may be:

3 Face : 1 Back (3F:1B)

- ❖ **Mixed Thread Arrangements**

Sometimes face and back threads are arranged in a mixed order, such as:

Warp: 1 Face, 1 Back

Weft: 2 Face, 1 Back

Warp: 2 Face, 1 Back

Weft: 2 Face, 2 Back

- ❖ **Irregular arrangements** are also used, for example:

5 Face : 4 Back

7 Face : 5 Back

These irregular arrangements allow **more flexibility** in choosing the amount of face and back threads.

CONSIDERING POINTS BEFORE DOUBLE CLOTH CONSTRUCTION

❑ Selection of Face and Back Yarn Thickness

✓ The thickness of face and back yarns should be selected according to the number of threads in the fabric.

✓ Examples:

❖ 1 Face : 1 Back Arrangement

- Back yarn should be similar or slightly thicker than the face yarn.

❖ 2 Face : 1 Back Arrangement

- Back yarn can be thicker than the face yarn.
- It may be about $\frac{2}{3}$ to $\frac{1}{2}$ thicker than the face yarn.

❖ When the same weave is used on both sides:

- In a 2 Face : 1 Back structure, back threads may be 3 - 4 times heavier than face threads, especially when center stitching threads are used.

CONSIDERING POINTS BEFORE DOUBLE CLOTH CONSTRUCTION

2. Selection of Face & Back Weave:

□ Equal Face and Back Threads

- ❖ When face and back threads are equal, the back weave is usually the same as the face weave.
- ❖ It may also have a similar number of yarn crossings (intersections).
- ❖ Example: 2/2 twill can be used as a back weave for 3/2 twill.

□ Unequal Face and Back Threads

- ❖ When the number of face and back threads are different, the back weave needs more intersections.
- ❖ This makes the back fabric stronger and balances the structure.
- ❖ Example (2 face : 1 back):
 - ✓ Plain weave can support 2/2 twill.
 - ✓ 2/1 twill can support 3/3 twill.
 - ✓ 2/2 twill can support 4/4 twill.

CONSIDERING POINTS BEFORE DOUBLE CLOTH CONSTRUCTION

❑ Fine Face and Soft Back Fabric

- ❖ For a fabric with a **smooth** face and soft back, **the same weave** can be used for both sides.
- ❖ In 1 face : 1 back fabric, the back weave can be looser than the face weave.

3. Tying or Stitching:

Stitching of the face & back fabrics of the double cloth can be affected in five ways. The first three methods are used for stitching. In two other methods the extra system either of warp or weft is introduced, which lies between the face and back fabric, stitching them.

➤ Stitching from Face to Back

- ❖ In this method, the **face warp goes below the back weft**.
- ❖ A face yarn connects the face fabric with the back fabric.
- ❖ The stitch becomes invisible when the yarn stays away from the outer surfaces.

CONSIDERING POINTS BEFORE DOUBLE CLOTH CONSTRUCTION

➤ Stitching from Back to Face

- ❖ In this method, the back warp goes above the face weft.
- ❖ A back yarn connects the back fabric with the face fabric.
- ❖ The stitch is invisible when the yarn does not appear on either surface.

➤ Combination Stitching

- ❖ Both methods are used together: Face-to-back stitching & Back-to-face stitching
- ❖ The warp threads of each fabric enter into the other fabric's structure.

➤ Stitching with Extra Warp

- ❖ An additional warp yarn is used to join the face and back fabrics.
- ❖ The face and back yarns do not interlace with each other.
- ❖ It uses: Three warp systems, Two weft systems
- ❖ Extra stitching warp should not be confused with extra stuffer warp.

CONSIDERING POINTS BEFORE DOUBLE CLOTH CONSTRUCTION

➤ Stitching with Extra Weft

- ❖ An extra weft yarn is used to join face and back fabrics.
- ❖ The fabrics are held together only by extra weft threads.
- ❖ Extra stuffing weft can be added to increase fabric thickness.
- ❖ If different types of extra weft yarns are used, a multi-shuttle loom is required.

4. The construction of the point paper design.

5. The beaming, drafting, the construction of the lifting plans, cross-section and longitudinal section.

SELECTION OF SUITABLE STITCHING POSITIONS

Importance of Stitch Placement

- A. In double cloth, the stitching points join the face and back fabrics together.
- B. If the stitches are placed correctly, they do not appear on either side of the fabric.
- C. The choice of stitching method depends mainly on the face weave structure.

Suitable Stitching Methods

1. Back Warp Raised (Back-to-Face Stitching)

- ❖ Suitable for: Warp satin, Warp-faced twill
- ❖ This method is usually preferred because the back warp is less likely to appear on the face side.

2. Face Warp Lowered (Face-to-Back Stitching)

- ❖ Suitable for: Weft sateen, Weft-faced twill
- ❖ In this method, the face warp goes below the back weft to join the two fabrics.

SELECTION OF SUITABLE STITCHING POSITIONS

Conditions for Perfect Stitching

When Back Warp is Raised, for an invisible stitch:

1. Back warp should not appear on the back surface.
2. It should pass through the correct position between face weave floats.
3. The face pick should not appear on the face surface.
4. The stitch point should be covered by back weft floats.

When Face Warp is Lowered, for an invisible stitch:

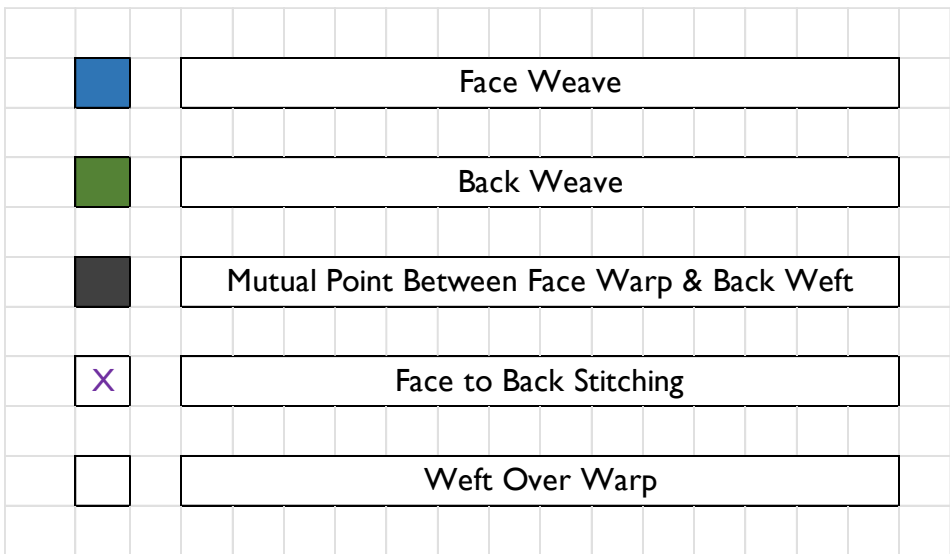
1. Face warp should not appear on the face surface.
2. It should be covered by back warp floats.
3. The back pick should not appear on the back surface.
4. The stitch should be covered by face weft floats.

SELECTION OF SUITABLE STITCHING POSITIONS

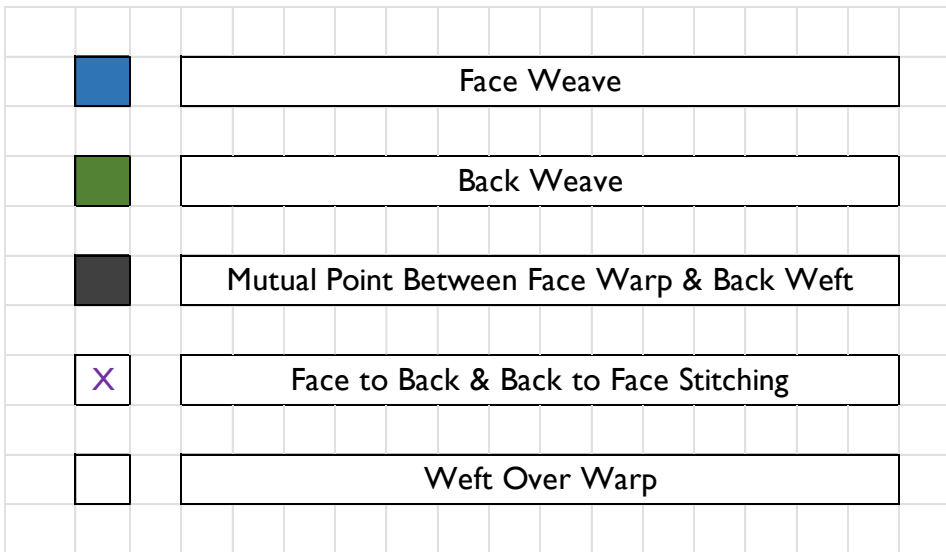
Main Points

- ❑ All conditions may not always be possible at the same time.
- ❑ The most important conditions are:
 - ❖ The stitching thread should not appear on the fabric surface.
 - ❖ The stitch should remain hidden.
 - ❖ Some compromise may be needed for perfect fabric appearance.

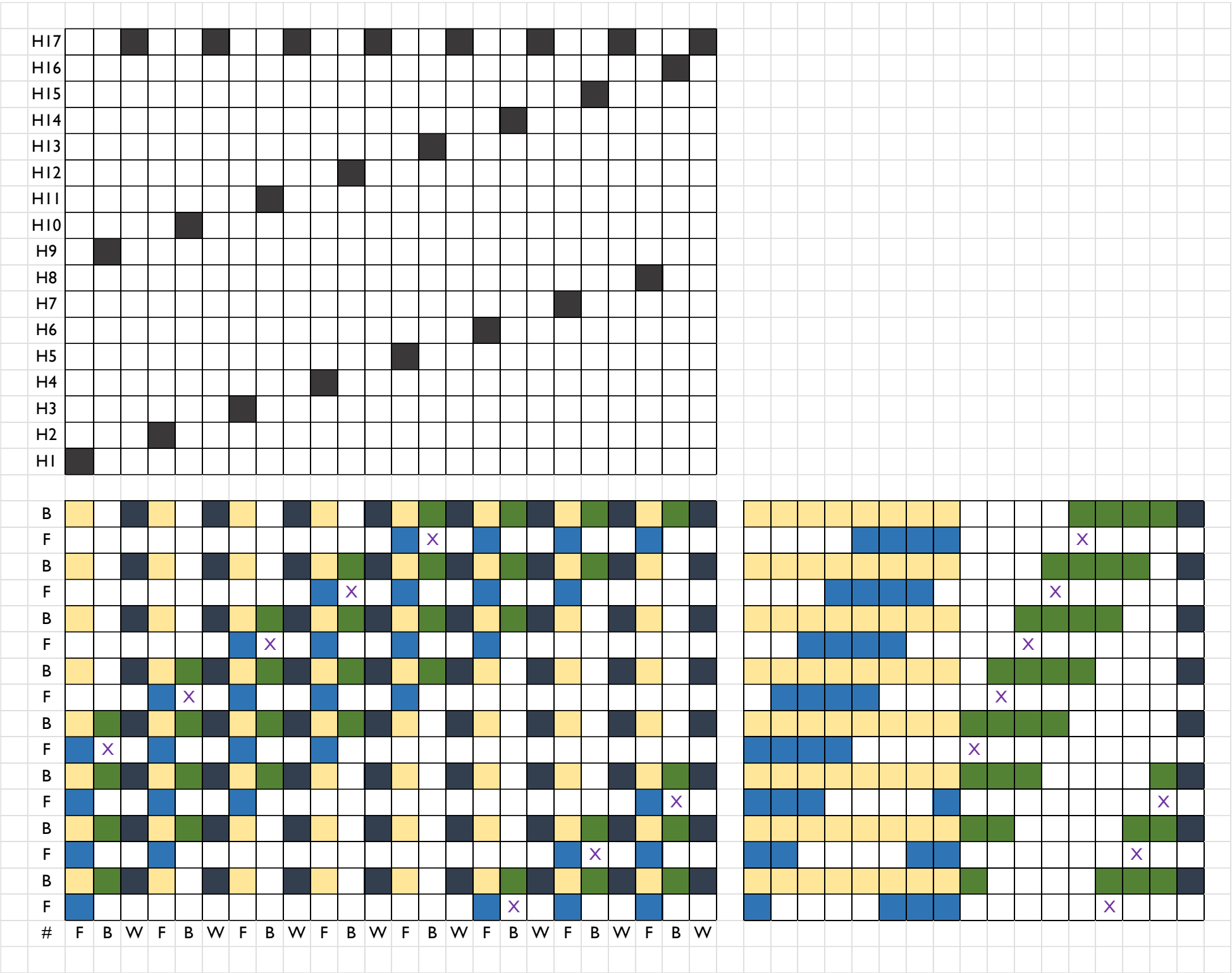
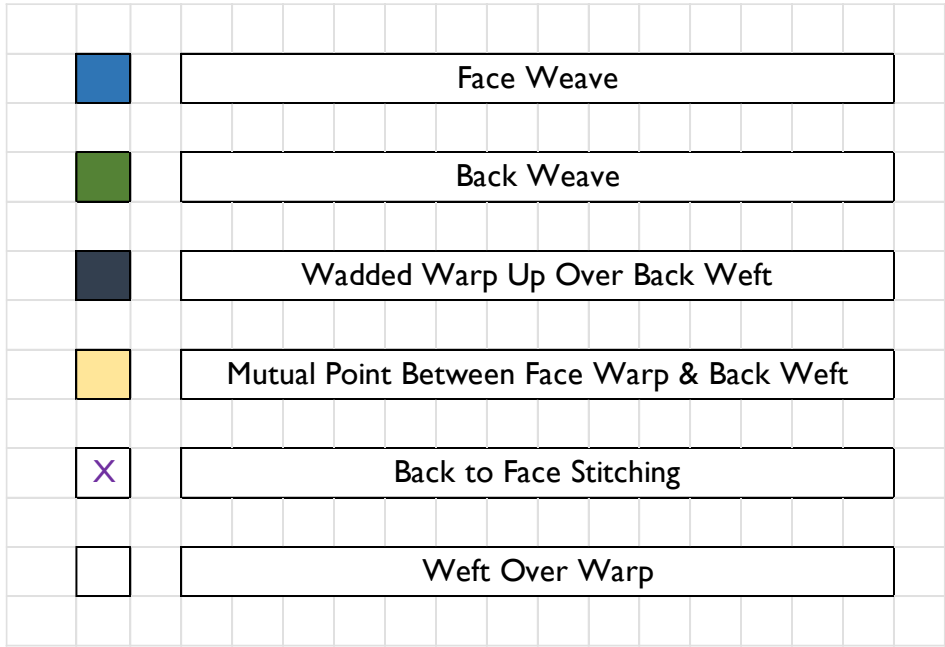
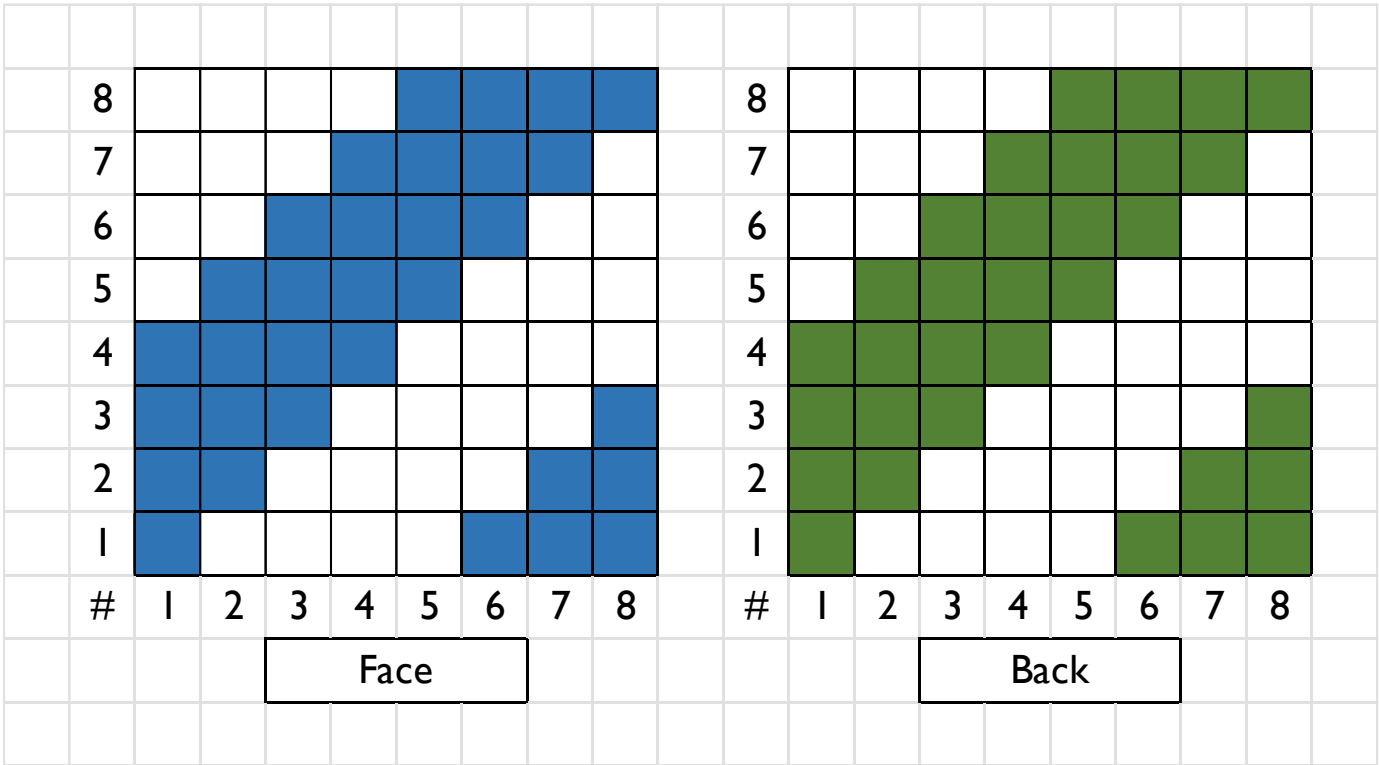
FACE TO BACK STITCHING



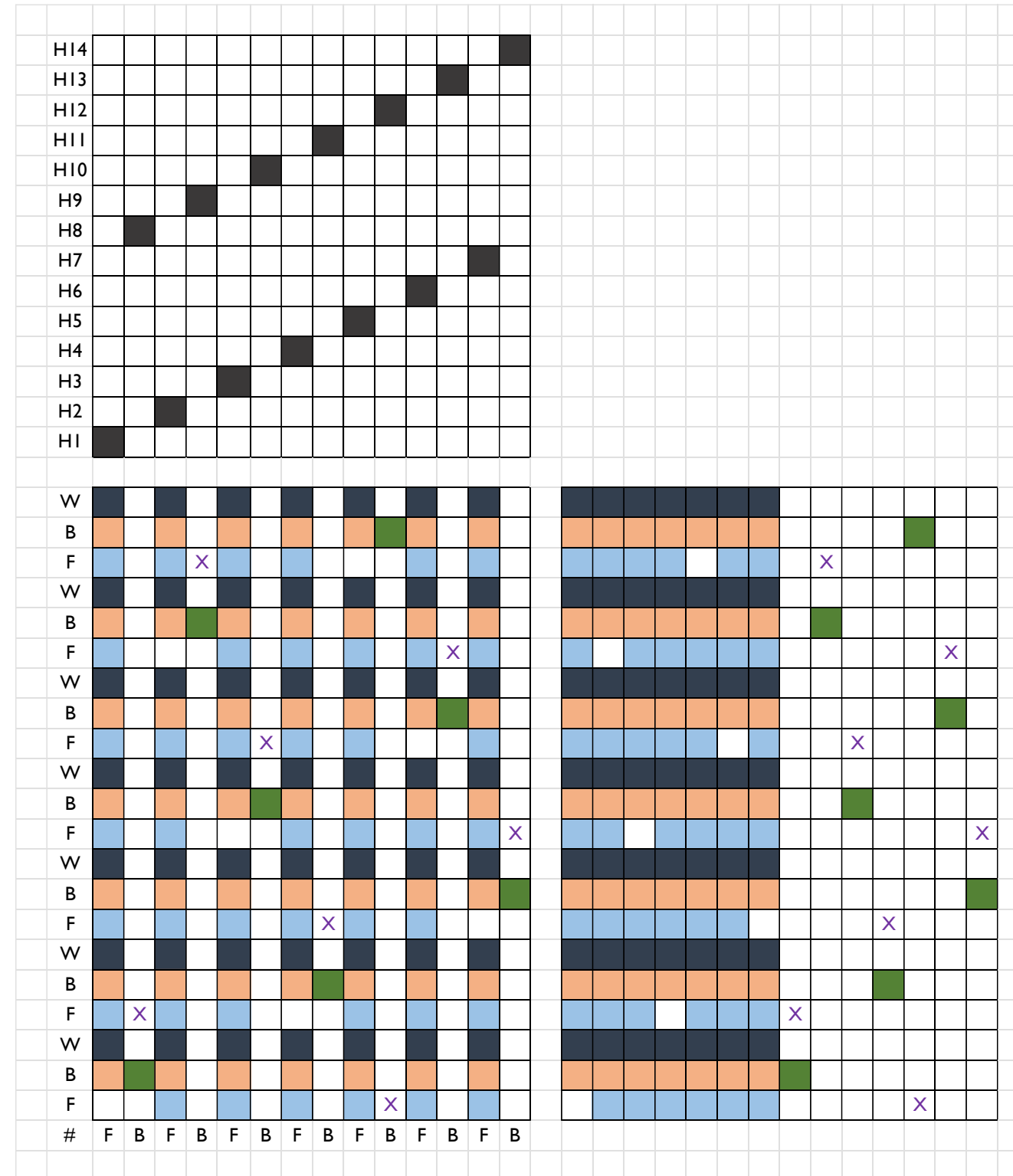
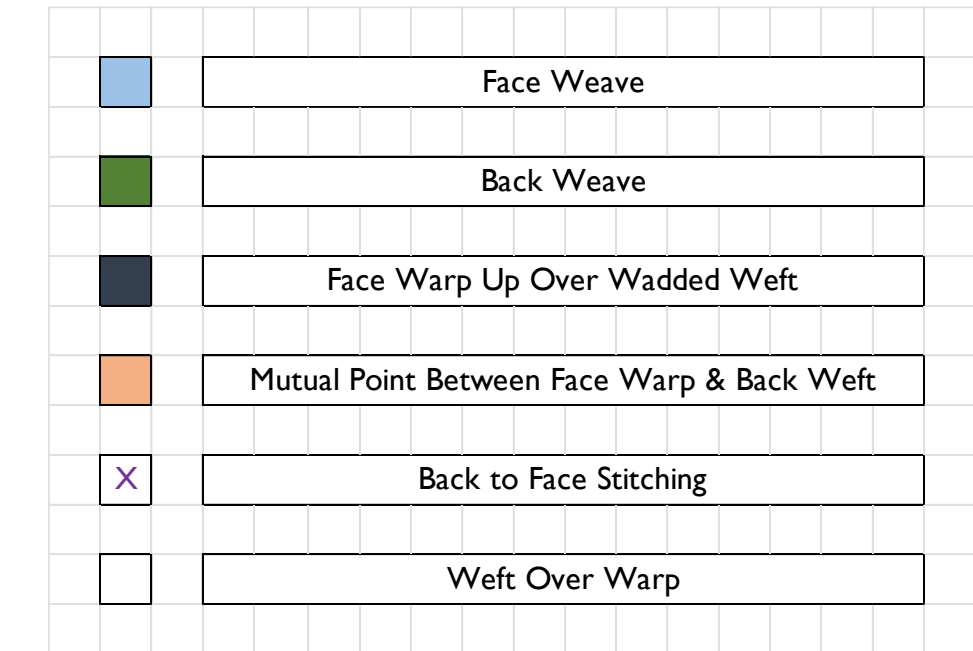
COMBINED STITCHING



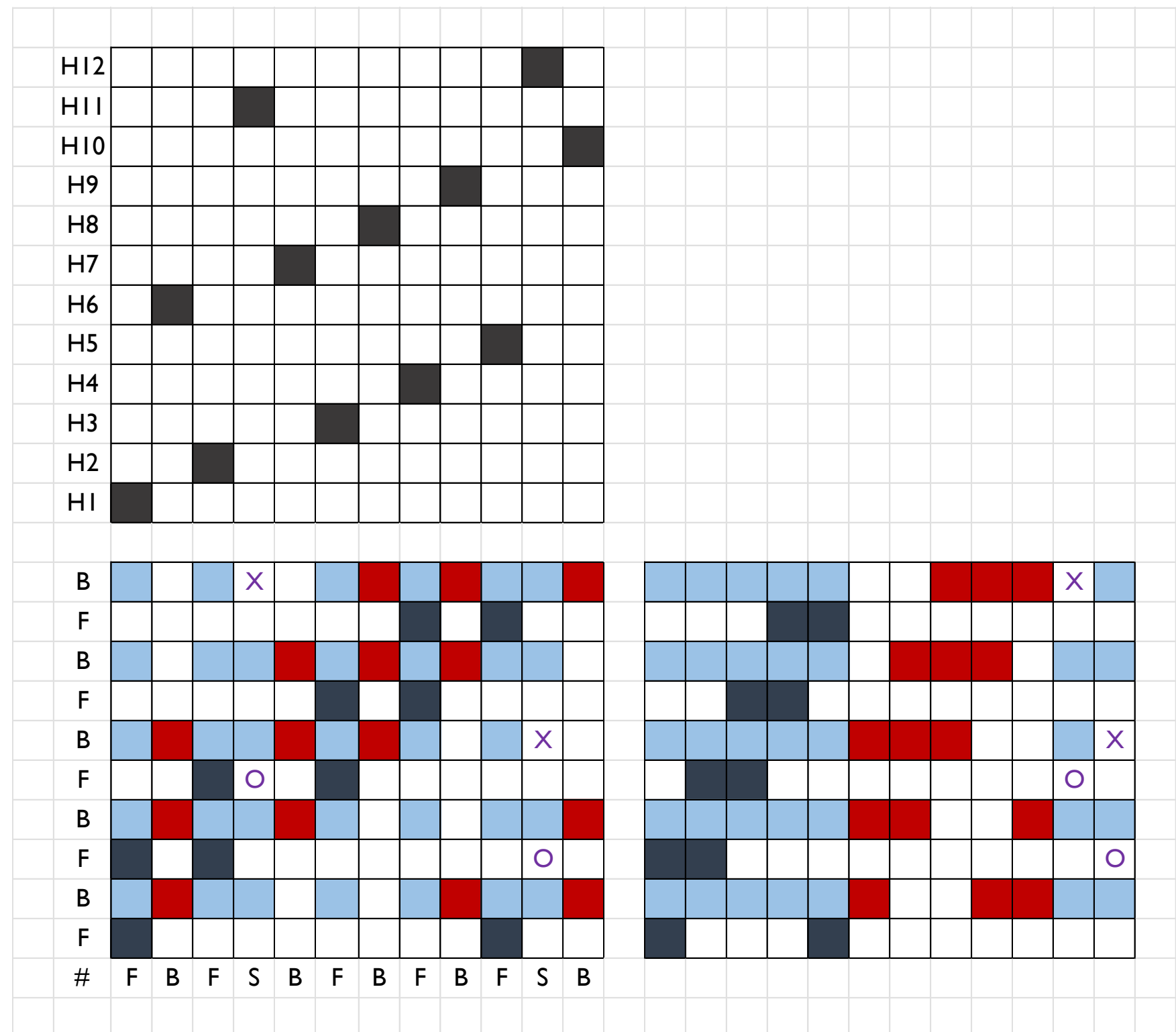
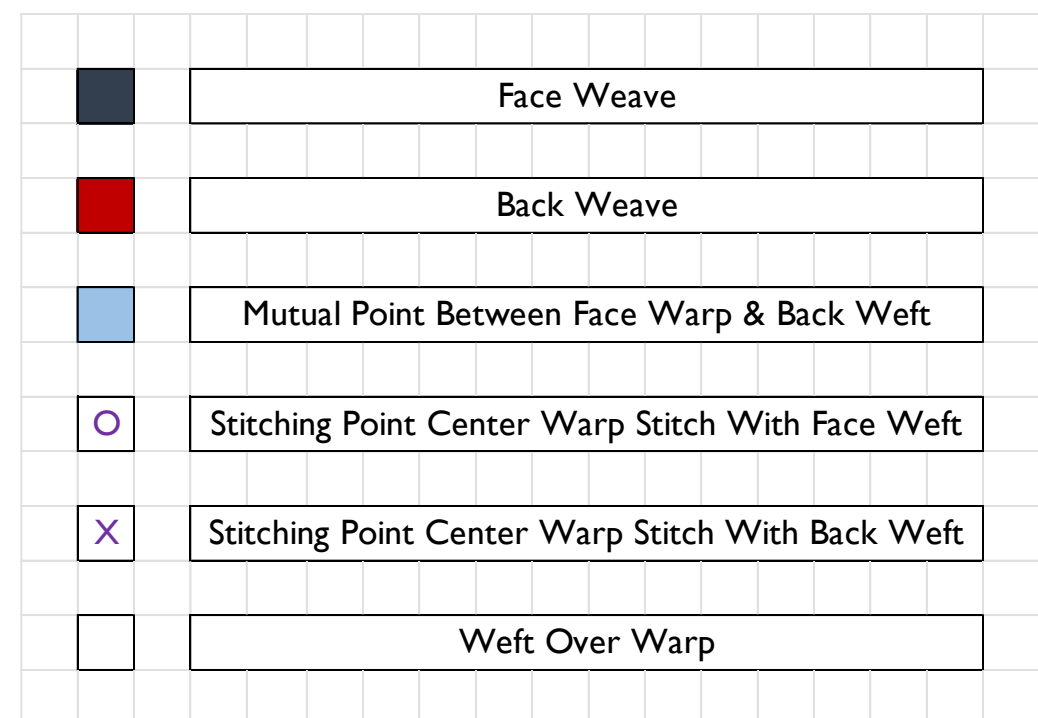
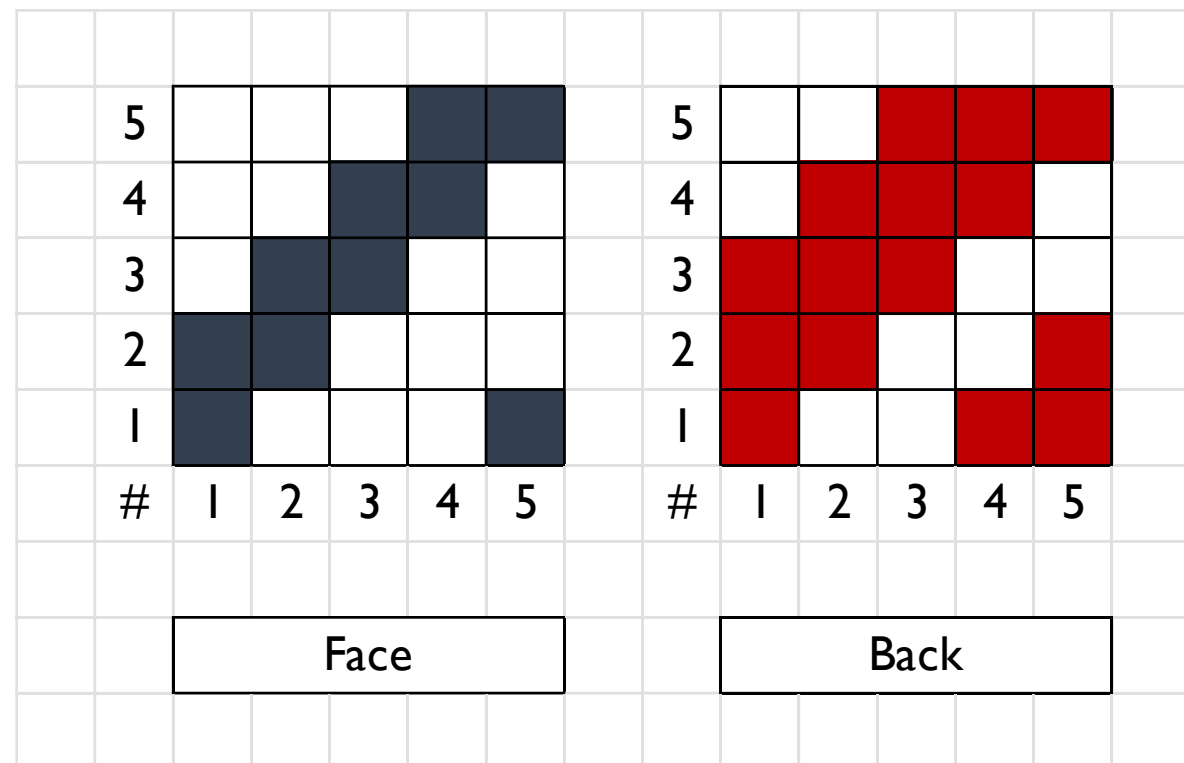
DOUBLE CLOTH BASED ON WARP WADDED STITCHING SYSTEM



WEFT WADDED STITCHING



CENTER WARP STITCHING



THANK
YOU !

